

Pokhara University
Gandaki College of Engineering and Science
Lamachaur, Pokhara-16, Kaski



Lab4: Implementation of Agile Project Management Tools
(JIRA, Asana, Trello, ClickUp, Zoho Projects).

Submitted By:

Sushan Tamang

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Roll no.41

Submitted To:

Er. Rajendra Bahadur Thapa

OBJECTIVE

Implementation of Agile Project Management using popular Agile Project Management Tools such as JIRA, Asana, Trello, ClickUp, and Zoho Projects.

AIM

To understand and implement Agile project management practices such as backlog creation, sprint planning, task tracking, and workflow visualization using various Agile Project Management tools, and to compare their features.

THEORY

Agile Project Management is an iterative approach to planning and guiding project processes, breaking down work into smaller, manageable units called sprints or iterations. It emphasizes collaboration, continuous feedback, and flexibility to adapt to changing requirements throughout the software development lifecycle.

Various software tools have been developed to support Agile practices such as Scrum and Kanban by providing features for backlog management, sprint planning, task boards, progress tracking, and team collaboration. Some of the widely used Agile Project Management tools are:

- JIRA – Developed by Atlassian, JIRA is one of the most widely used Agile tools, offering Scrum and Kanban boards, backlog management, sprint planning, and detailed reporting (burndown charts, velocity charts).
- Asana – A work management tool that allows teams to organize tasks into projects, lists, boards, and timelines, with features for task assignment, due dates, and dependencies.

- Trello—A simple, visual, Kanban-style tool that uses boards, lists, and cards to organize tasks, making it easy to track the progress of work items across different stages.
- ClickUp – An all-in-one productivity and project management platform that combines task management, docs, goals, and time tracking with support for multiple views (List, Board, Gantt, Calendar).
- Zoho Projects—A cloud-based project management tool by Zoho that offers task management, Gantt charts, time tracking, and collaboration features, integrated with other Zoho applications.

These tools help Agile teams plan sprints, visualize workflow, track progress in real time, and improve collaboration and transparency, which are essential principles of Agile Software Development.

OBJECTIVES

- Learn to create and manage projects, boards, and tasks in different Agile tools.
- Understand backlog creation and sprint planning using JIRA.
- Understand task and project organization using Asana.
- Understand Kanban-style workflow visualization using Trello.
- Explore multi-view task management using ClickUp.
- Explore project tracking and reporting using Zoho Projects.

1. AGILE PROJECT MANAGEMENT USING JIRA

JIRA is an Agile project management tool widely used by software teams to plan, track, and manage software development projects using Scrum or Kanban methodology.

Algorithm

1. Create a JIRA account and log into the JIRA dashboard.
2. Create a new project and select the Scrum template.
3. Create a Product Backlog by adding user stories/issues (e.g., 'As a user, I want to log in').
4. Estimate story points for each backlog item.
5. Create a new Sprint and move selected backlog items into it.
6. Start the sprint and assign tasks to team members.
7. Move tasks across the board (To Do -> In Progress -> Done) as work progresses.
8. Complete the sprint and review the Burndown Chart.

Sample Backlog/ Sprint Setup

Project: E-Commerce Website

Sprint 1 (2 weeks) - Backlog Items:

ECOM-1 As a user, I want to register an account	[5 pts]
ECOM-2 As a user, I want to log in securely	[3 pts]
ECOM-3 As a user, I want to view product listings	[8 pts]
ECOM-4 As a user, I want to add products to cart	[5 pts]

Output (Sprint Board Status)

SPRINT1 BOARD

TO DO | IN PROGRESS | DONE

ECOM-4 | ECOM-2 | ECOM-1
 | ECOM-3 |

Sprint Progress: 40% complete (8/21 story points done)

Burndown Chart: On track

2. AGILE PROJECT MANAGEMENT USING ASANA

Asana is a work and project management tool that allows teams to organize tasks into projects, assign owners, set due dates, and track progress using List, Board, or Timeline views.

Algorithm

1. Create an Asana account and log into the dashboard.
2. Create a new Project (e.g., 'Website Redesign').
3. Choose a view: List, Board (Kanban), or Timeline.
4. Add sections to represent workflow stages (To Do, Doing, Done).
5. Create tasks under each section and add details (assignee, due date, priority).
6. Add subtasks and dependencies where required.
7. Update task status as work progresses and mark tasks complete.
8. Monitor overall project progress from the Project Overview/Dashboard.

Sample Task List

```
Project: Website Redesign
Section: To Do
- Design homepage wireframe      (Assignee: Samika, Due: 2 days)
- Collect client requirements     (Assignee: Team, Due: 1 day)
Section: In Progress
- Develop navbar component       (Assignee: Samika, Due: 3 days)
Section: Done
- Setup project repository       (Completed)
```

Output (Project Progress Dashboard)

```
Website Redesign - Progress Overview
-----
Total Tasks: 4
```

Completed:1 (25%)

InProgress:1 (25%)

ToDo: 2(50%)

Status: Ontrack,no overdue tasks

3. AGILE PROJECT MANAGEMENT USING TRELLO

Trello is a lightweight, visual Kanban-style project management tool that uses boards, lists, and cards, making it simple to track the movement of tasks across different workflow stages.

Algorithm

1. Create a Trello account and log in.
2. Create a new Board (e.g., 'Mobile App Development').
3. Create Lists representing workflow stages: Backlog, To Do, In Progress, Testing, Done.
4. Add Cards under each list representing individual tasks.
5. Add details to cards: labels, due dates, checklists, and assigned members.
6. Move cards across lists (drag-and-drop) as tasks progress.
7. Use labels/colors to indicate priority or task type.
8. Review the board regularly to monitor overall progress.

Sample Board Structure

Board: Mobile App Development

Backlog: [Design splash screen] [Setup Firebase]

To Do: [Implement login UI]

In Progress: [Implement API integration]

Testing: [Test login flow]

Done: [Project setup] [Wireframe approved]

Output(BoardSnapshot)

MOBILEAPPDEVELOPMENTBOARD

BACKLOG(2)|TODO(1)|INPROGRESS(1)|TESTING(1)|DONE(2)

Cardsmovedthisweek:3

Overdue cards: 0

Boardactivity:Active

4. AGILEPROJECTMANAGEMENTUSINGCLICKUP

ClickUp is an all-in-one productivity platform that supports multiple task views (List,Board,Gantt,Calendar)alongwithgoals,docs,andtimetracking,making it suitable for both Agile and traditional project management.

Algorithm

1. CreateaClickUpaccountandlog into theworkspace.
2. Createa newSpace andList/Project(e.g.,'MarketingCampaign').
3. AddTaskswithdetailssuchaspriority,assignee,duedate,andtags.
4. Organizetasksusingstatuses(Open,InProgress,Review,Complete).
5. SwitchbetweenBoardview(Kanban)andGantt/Calendarviewsas needed.
6. SetGoals/Milestonetotracksprintorproject objectives.
7. Usethe Time Trackingfeature tolog hours spentontasks.
8. ReviewtheDashboard for anoverviewof workloadand progress.

SampleTaskSetup

Space: Marketing Team

List:Q3CampaignLaunch

Tasks:

-Createsocial mediacontent [Priority:High] [Status:In Progress]

- | | |
|-------------------------|-------------------------------------|
| - Designemailnewsletter | [Priority: Normal] [Status: Open] |
| - Scheduleadcampaign | [Priority:Urgent][Status:Review] |
| - Publishlandingpage | [Priority: High] [Status: Complete] |

Output(WorkloadDashboard)

Q3CampaignLaunch- StatusSummary

Open: 1 In Progress: 1 Review: 1 Complete:1

Total time logged this week: 18h 30m

GoalProgress:60%(3/5milestonesachieved)

5. AGILE PROJECT MANAGEMENT USING ZOHO PROJECTS

Zoho Projects is a cloud-based project management tool that supports task management, Gantt charts, time tracking, and team collaboration, and integrates seamlessly with other Zoho applications.

Algorithm

1. CreateaZoho accountand login to Zoho Projects.
2. Createa newProject(e.g.,'ERPSystemDevelopment').
3. DefineTaskLists/Modules andaddindividualTasksundereach.
4. Assigntaskstoteam membersandset start/end dates.
5. Set taskdependenciesandviewthem ontheGanttChart.
6. Logworkhoursusingthebuilt-inTimesheet/Timerfeature.
7. Trackoverall projectstatususingtheProjectDashboard.
8. Generatereportsto reviewtaskcompletion and teamperformance.

SampleTaskList

Project: ERPSystemDevelopment

Task List:Module-InventoryManagement

- Design database schema (Start:Day1,End: Day3)
- Develop inventory API (Start:Day4,End: Day8)
- Integrate with frontend (Start:Day9,End: Day12)
- Perform module testing (Start:Day13,End:Day15)

Output(ProjectDashboardSummary)

ERP System Development- Dashboard

TotalTasks:4 Completed:1 Overdue:0

ProjectCompletion:32%

Hours Logged: 42h

Status:On schedule

RESULT

Agile project management practices such as backlog creation, sprint planning, task assignment, and progress tracking were successfully implemented and demonstrated using JIRA, Asana, Trello, ClickUp, and Zoho Projects.

CONCLUSION

This experiment provided hands-on exposure to widely used Agile Project Management tools. JIRA proved effective for structured Scrum-based sprint planning and reporting, Asana and ClickUp offered flexible task and workflow management with multiple views, Trello provided a simple and visual Kanban-based approach, and Zoho Projects combined task tracking with Gantt charts and time management. Overall, these tools enhanced team collaboration, transparency, and efficiency, which are core principles of Agile Software Development.